

Nifty ORB Alert System

Operator Handbook - V5

Risk-sized 1-lot | Auto exits | Fill-confirmed | Conviction tag

June 2026

Nifty ORB Alert System — Handbook (V7)

Your daily operating manual for the breakout alert + 1-tap order bot running on AWS Lightsail.

> STATUS — LIVE from Mon 29 Jun 2026. After validating the strategy end-to-end (the UP/CE > monthly setup reproduces ~11.7% net OOS CAGR, Sharpe ~1.7) and fixing the exit to the > validated stepped trailing stop, the bot is going LIVE with real money, 1 lot per > signal, for a 3-month trial. DRY_RUN=False, PAPER_TRADE=False, PAPER_AUTO_ENTER=False > (you tap each BUY yourself). Goal of week 1: shake out real order-placement issues. Worst-case > risk per trade is ~Rs.9,750 (1 lot, 30% stop, premium ≤Rs.500). Weekly/early-cycle signals > stay paper-only and are never real money.

What this system does

Every trading day the bot watches the Nifty current-month futures on Zerodha and alerts you on Telegram when price breaks cleanly out of the first 15-minute range (the "ORB"). A qualifying breakout arrives with a one-tap BUY button for an ITM-1 option. You tap to enter; the bot exits on its own (a -30% stop that *ratchets up in 15% steps* to lock gains as the trade rises — "let winners run" — or the end-of-day square-off). It only trades real money once you set PAPER_TRADE=False and DRY_RUN=False.

New in V7 — you only trade the last ~2 weeks of the monthly cycle. The bot now picks the option *expiry* by how close the monthly expiry is: in the last 15 days before monthly expiry it offers an actionable MONTHLY call (the trades you take); earlier in the cycle it offers a weekly call tagged TRACKING ONLY — recorded as paper to measure performance, with no button and no real money. See "Which option you'll be offered" below.

Who does what:

- > You do: the morning Kite login, and one button tap for each trade you choose to take.
- > The bot does: watches the market, applies every filter, sizes the trade, places the order, and manages all exits automatically.
- > Built-in protection: fixed 1 lot per trade (never scales up), options priced above Rs.500 are skipped (worst-case ~Rs.9,750 risk/trade at a 30% stop), and a Rs.15,000 monthly loss limit pauses new trades for the rest of the month if hit.

How the system decides (the full pipeline)

A BUY recommendation appears only when every check below is YES. Each one is a noise filter — a single NO means no alert.





The same filters in words: 1. ORB formed — the 9:15–9:30 candle sets the day's High and Low. 2. Open-inside-range — the breakout candle must *open inside* the ORB (no chasing a move that already ran away). 3. Clean break — close must clear ORB High +10 pts (UP) or ORB Low -10 pts (DOWN). 4. Volume — breakout volume > 1.2× the 20-candle average. 5. VWAP — close above VWAP for UP, below for DOWN. 6. Value area (VAH/VAL) — the close must clear the day's value area: above VAH for UP, below VAL for DOWN. (This used to be only a label; it's now a real filter, matching the backtest.) 7. Option-OI confirmation — the open-interest of the two out-of-the-money options *against* the move must have risen since 9:30:

- > UP → the two puts just below spot (put writers building support = bullish confirm). This is the put-side OI edge that passed the walk-forward backtest.
- > DOWN → the two calls just above spot (call writers building resistance overhead = bearish confirm). This is the mirror gate added for the PE side. (!) Note: this is not a validated edge — the PE side lost money in backtest, and call-side OI was rejected on the CE side; it's here as a noise filter, which is why every DOWN alert is flagged LOW conviction (see below).

If the 9:30 reading can't be fetched, the check is skipped for that day (fail-open) so a data hiccup doesn't block

everything. 8. Armed / fire limit — max 2 fires per direction per day; after a fire, price must pull back ≥ 15 pts into the range to re-arm. 9. Expiry + option — pick ITM-1 on the monthly (≤ 15 days to monthly expiry $\rightarrow$ actionable) or the weekly (earlier in the cycle $\rightarrow$ tracked only); skip the button if spread $> 5\%$ or premium $> \text{Rs.}500$. 10. Risk sizing — take 1 lot only if its worst-case stop-loss is $\leq \text{Rs.}5,000$ (premium $\leq \text{Rs.}256$); otherwise skip. 11. Monthly loss limit — if the month's realized loss has hit $\text{Rs.}15,000$, pause until next month.

Because UP breakouts must now clear VAH to fire at all, every UP alert is high-conviction by construction (you'll still see the ★ marker on it). DOWN breakdowns now pass the same shape of gate (VAL + rising call-OI), but because the PE side is not a validated edge every DOWN alert is flagged (!) LOW CONVICTION (PE — experimental, paper only) — surface it, don't bet real money on it.

After you tap: the bot runs the trade by itself



You do not tap to exit — exits are automatic. In live mode, both entry and exit are fill-confirmed: if a BUY doesn't fill because price jumped, you get "BUY NOT FILLED" and no position is taken.

Which option you'll be offered: weekly (track) vs monthly (act)

The bot chooses the option expiry from how many days are left to the monthly expiry — and only the monthly half is yours to trade.

Last ~2 weeks of the cycle $\rightarrow$ MONTHLY (you ACT). When the monthly contract is 15 days or fewer from expiry, the bot recommends the monthly ITM-1 call, tagged ACTIONABLE with a BUY button. These are the trades you take. This is the validated DTE-monthly strategy.

Earlier in the cycle $\rightarrow$ WEEKLY (TRACKING ONLY, you do nothing). When the monthly is more than 15 days out, the bot recommends a weekly ITM-1 call but tags it TRACKING ONLY — no button, and it's recorded as a *paper* trade purely to measure how the weekly half performs. These stay paper even after you go live — never real money. You ignore them.

Every alert tells you which half you're in:

- > Expiry: MONTHLY · 4d to expiry — ACTIONABLE → a real recommendation (trade it if you choose)
- > Expiry: WEEKLY · monthly 22d out — TRACKING ONLY (no order) → ignore; it's only being recorded

In one line: in the ~2 weeks before each monthly expiry you get real monthly recommendations to act on; the rest of the month the bot quietly paper-tracks weeklies and you sit out. The switch is automatic — the cutoff is DTE_MONTHLY_MAX (default 15) in config.py, and WEEKLY_TRACK_ONLY=True keeps the weekly half paper-only.

Every day: your 4 steps

This is the entire daily routine — about a minute.

1. ~8:30 AM — you get a Telegram nudge with the Kite login link.
2. Log in to Kite (Zerodha + 2FA) and copy the redirect URL from the browser.
3. Cache today's token on the VM:

```
ssh orb-vm
cd ~/nifty-orb-alerts
python3 auth.py          # paste the URL when prompted
```

4. Start the bot:

```
./start.sh
```

You'll get an "ORB monitor started" message on Telegram. Done — close SSH; the bot keeps running.

During the day: when a breakout alert arrives, tap BUY if you want that trade, or ignore it. The bot handles the stop-loss, target, and 15:15 square-off automatically. To stop the bot early: ./stop.sh.

Pre-flight: the DRY_RUN flag

The bot ships with DRY_RUN = True in config.py. In this mode:

- > All alerts and buttons appear normally on Telegram
- > When you tap the button, the bot logs the order it would have placed and confirms back: [DRY_RUN] Order placed: NIFTY26MAY24450CE BUY 65 @ LIMIT Rs.152.95
- > No real money moves

Run for one full trading day in DRY_RUN. Verify each alert: is the symbol right (correct expiry, correct CE/PE), is the strike one ITM (50 pts in the money for Nifty), is the price near the visible Kite quote? Once you've seen 2-3 clean alerts pass that check, flip the flag:

```
ssh ubuntu@<lightsail-ip>
cd ~/nifty-orb-alerts
nano config.py          # change DRY_RUN = True -> DRY_RUN = False
pkill -f orb_monitor.py
nohup python orb_monitor.py > orb.log 2>&1 &
```

The startup Telegram message will now say ORB monitor started (LIVE) instead of (DRY_RUN).

Paper trading mode (V3)

Before flipping DRY_RUN = False (which uses real money), run PAPER_TRADE = True for a couple of weeks. Paper

mode is the realistic dress rehearsal:

- > All alerts fire normally on Telegram with BUY buttons.
- > When you tap the button, no Kite order is placed — but the bot:

1. Journals the trade in trades.db at the alert-time LIMIT price (same value LIVE will use). 2. Starts monitoring the position every minute (POSITION_POLL_SECONDS = 60). 3. Auto-exits on the trailing stop (initial -30%, then trails up to lock gains; exit reason TRAIL) or TIMEOUT (15:15 IST square-off). *(With USE_TRAILING_STOP=False it reverts to the old fixed -30% SL / +50% TARGET.)* 4. Sends a Telegram exit alert with realized P&L.

This builds a full P&L history without risking capital. Mode truth table:

'PAPER_TRADE'	'DRY_RUN'	Mode	Real order?	DB row?	Auto-exit?
True	True	PAPER	No	Yes	Yes
True	False	PAPER	No	Yes	Yes
False	True	DRY_RUN	No	No	No (alert-only)
False	False	LIVE	Yes	Yes	Yes

Important: PAPER_TRADE overrides DRY_RUN. To go live, set both to False.

How the journal records prices

The journal records the price from the alert (e.g., LIMIT: Rs.152.95 shown on the BUY button), not the live ask at the moment you tap. This means:

- > In LIVE, your Kite LIMIT order is placed at exactly that price — fills land at or near it.
- > In PAPER, the same price is recorded — so the journal measures signal quality, not tap-reaction speed.

If you tap a button 5 minutes late and the option has moved Rs.30, the journal still uses the original alert price. The position monitor uses real-time LTP for SL / target / timeout decisions, so the trade is still evaluated against current market — but the entry P&L is anchored to the price the bot recommended.

Configurable exit levels (in config.py)

Setting	Default	Meaning
SL_PCT	0.30	Initial stop 30% below entry
USE_TRAILING_STOP	True	Stepped trailing stop, no fixed target ("let winners run"). False = old fixed SL/target
TRAIL_STEP_PCT	0.15	Ladder step: the stop ratchets up 15% of entry for every 15% the option gains. Matches the validated backtest
TARGET_PCT	0.50	Fixed target — ignored when USE_TRAILING_STOP=True
SQUARE_OFF_HOUR	15	Hard square-off hour (IST)
SQUARE_OFF_MINUTE	15	Hard square-off minute. 15:15 = 15 min before close
POSITION_POLL_SECONDS	60	How often to poll the option for exit conditions

RISK_PER_TRADE_INR	15000	Sized so 1 lot always qualifies up to the premium cap (see below)
MAX_LOTS	1	Hard ceiling on lots per trade — never sizes beyond one lot
MAX_PREMIUM	500	Options priced above this are skipped (no button); caps worst-case risk
MONTHLY_LOSS_LIMIT_INR	15000	Month's realized loss cap; pauses new trades when hit

How the stepped stop works (validated). The stop opens at -30% of entry and then ratchets up in 15% steps as the option climbs: it only lifts after each full +15% gain, and holds between steps instead of tightening on every tick. Example (entry Rs.100): stop starts at Rs.70; once the option touches Rs.115 the stop steps to Rs.85; at Rs.130 it steps to Rs.100 (breakeven); at Rs.160 → Rs.130; and so on. This is what lets winners run — a continuous "always 30% below the peak" trail backtested ~1/3 worse (11.7% → 7.5% net), which is why the step matters.

Position sizing & the monthly loss limit (V4)

Two guards now bound how much you can lose. Both are automatic — you don't tap anything extra.

1. Fixed 1 lot per signal, capped by premium. For the live trial the bot trades a fixed one lot (65) on every actionable signal — matching the validated backtest, which was always 1 lot. RISK_PER_TRADE_INR is set to Rs.15,000 purely so the risk-sizer never rounds one lot down to zero; with MAX_LOTS = 1 it can never scale you *up* past one lot.

The only thing that skips a trade is the premium cap (MAX_PREMIUM, Rs.500): options priced above Rs.500 get an alert but no button (place manually only if you truly want it). The worst case on a taken trade is therefore:

> one-lot risk = option premium × SL_PCT (0.30) × LOT_SIZE (65) = premium × 19.5 > → at the Rs.500 cap, worst-case loss ≈ Rs.9,750 for one lot.

Why one lot only? MAX_LOTS = 1 is a hard ceiling so sizing can only ever *skip* a trade, never scale you into a bigger position — deliberate downside protection during the live trial.

2. Monthly loss limit (MONTHLY_LOSS_LIMIT_INR, default Rs.15,000). The bot adds up your realized (closed) P&L for the current calendar month. Once losses for the month reach Rs.15,000, it stops offering new trades for the rest of the month — you'll see monthly loss limit hit. It resets on its own at the start of the next month.

-> Any position already open is still managed normally (stop-loss, target, square-off all run). The limit only blocks *new* entries.

-> During paper trading the limit counts paper P&L; once live, it counts live P&L.

These two are your hard guardrails. Treat a skip or a pause as the system doing its job — the whole point is to make a single bad trade, or a bad month, impossible to turn into a disaster.

Daily morning ritual

You'll get two Telegram messages from the system before market open:

8:30 AM IST — Token refresh nudge

> CRITICAL TIMING — log in only AFTER ~7:45 AM IST. Zerodha invalidates all > access tokens at its morning reset (~7:30 AM). A token generated *before* the reset > works for a few minutes (the bot will even start fine) but is silently killed at > 7:30, and the bot then crashes at the 9:30 ORB fetch with TokenException — exactly > what

happened on 2026-06-29 (logged in at 05:20). Wait for this 8:30 nudge, then log > in. Never run auth.py before ~7:45. (V6.2 now also re-checks the token at 9:10 and > alerts instead of crashing, but the fix is to log in late.)

A message like:

```
Good morning. Wed, 30 Apr 2026

Refresh Kite access token before 9:15 IST.

1. Tap to log in:
https://kite.zerodha.com/connect/login?api_key=XXX&v=3

2. SSH to Lightsail and run:
cd ~/nifty-orb-alerts
python3 auth.py
./start.sh
```

Steps to follow (~60 seconds)

1. Tap the Kite login link in the Telegram message → log in with Zerodha + 2FA. 2. After login, you land on a page that may look broken — that's fine. Copy the full URL from the browser address bar (it contains request_token=...). 3. SSH to Lightsail from your laptop or AWS browser-based SSH:

```
ssh orb-vm                # or: ssh ubuntu@<lightsail-ip>
cd ~/nifty-orb-alerts
python3 auth.py
```

4. Paste the URL when prompted. You'll see Access token saved for today. 5. Start the monitor (this stops any old process and starts a fresh one):

```
./start.sh
```

start.sh first checks today's token is cached, then restarts the bot. If the token is missing it tells you to run python3 auth.py instead of failing silently. To stop the bot any time: ./stop.sh. 6. Within a minute, you should get a Telegram alert: ORB monitor started (LIVE) (or (PAPER)). If yes — you're done. Close the SSH window. The bot keeps running.

> Watchdog (V6.2): a cron job (healthcheck.py, every 10 min during market hours) pings you (!) WATCHDOG: orb_monitor is NOT running if the process ever disappears mid-session — so a silent death can't go unnoticed all day again. It alerts over Telegram, which is itself occasionally flaky on the VM, so treat a *missing* heartbeat as a prompt to glance at the bot, not a guarantee.

Why the bot keeps running after you close SSH

This is the part most people ask about, so here's the plain explanation:

- > nohup (= "no hangup") tells Linux: "if my session ends, don't kill this process."
- > The trailing & puts it in the background so the shell prompt comes back immediately.
- > > orb.log 2>&1 sends both normal output and errors into the file orb.log so the process doesn't try to write to a terminal that's gone.

Net effect: you can close the SSH window, close your laptop lid, fly across the country — the bot keeps running on the Lightsail VM until either:

- > You explicitly kill it (see "Daily shutdown" below), or

-> The VM itself reboots (rare; AWS usually warns you).

If your VM does reboot, the bot will not restart automatically — see the optional systemd setup at the end of this handbook to fix that.

What the alerts mean

"ORB monitor started"

Confirmation the bot is alive. Includes mode (DRY_RUN/LIVE), futures symbol, lot size. No action.

"ORB formed" (around 9:30:30 AM)

```
ORB formed for NIFTY24MAYFUT
High: 22687.40
Low: 22641.25
Buffer: 10 pts | Re-arm: 15 pts
Watching 5-min closes for breakout/breakdown...
```

Your day's range is set. Wider ranges (>80 pts) often mean choppy days — be more selective.

"BREAKOUT (UP) — fire 1/2" — the real signal with a button

```
BREAKOUT (UP) -- fire 1/2
NIFTY24MAYFUT 5m close: 22693.10 > ORB High 22687.40 (+10 buf)
Vol: 142500 (avg 89200, x1.60)
VWAP: 22669.85
Spot Nifty: 22691.55
Expiry: MONTHLY · 4d to expiry -- ACTIONABLE
ITM-1 CE: NIFTY24MAY22650CE
Bid/Ask: 152.10/152.45
LIMIT: Rs.152.95 | Qty: 65 (risk <= Rs.5000)

[ BUY 65 CE @ Rs.152.95 ] <- tap to place order
```

-> fire 1/2 = first breakout in this direction today

-> x1.60 = breakout candle volume was 1.6× the 20-candle avg — strong

-> Expiry: MONTHLY ... ACTIONABLE = you're in the last ~2 weeks of the monthly cycle, so this is a real recommendation with a button. (A WEEKLY ... TRACKING ONLY line means ignore it — see below.)

-> ITM-1 CE = 50 pts in the money (better delta than ATM for short holds)

-> Bid/Ask = the live option quote at the moment the alert fired

-> LIMIT Rs.152.95 = best_ask + 0.50 rounded to 0.05 tick — almost guaranteed to fill at the visible ask

Tap the button → bot places a BUY LIMIT order via Kite → confirmation message comes back:

```
Order placed: NIFTY24MAY22650CE BUY 65 @ LIMIT Rs.152.95
Order ID: 250502000123456
```

If the order fails (margin, rejected, network), you'll see:

```
ORDER FAILED for NIFTY24MAY22650CE: <error message from Kite>
```

...and you can place the trade manually in Kite.

"BREAKDOWN (DOWN) — fire 1/2"

Same shape but opposite — buys an ITM-1 PE (50 pts above spot). Every DOWN alert now carries a (!) LOW CONVICTION (PE — experimental, paper only) line. The PE side passes the mirror gate (closed below VAL + resistance-call OI rising since 9:30) but is not a validated money-maker — treat these as paper/observation only.

"fire 2/2"

A re-entry breakout after price came back inside ORB by ≥ 15 pts. Often cleaner than fire 1 because the noise has been shaken out.

"Auto-order skipped" (any reason)

You'll see this if the safety guards rejected the auto-order:

- > spread 7.2% > 5% — option spread too wide; place manually with a tight limit
- > premium Rs.620 > MAX_PREMIUM Rs.500 — option too expensive (often deep ITM after a big move)
- > quote fetch failed: ... — Kite API hiccup; refresh and place manually
- > 1-lot risk Rs.5640 > risk cap Rs.5000 — one lot of this option would risk more than your per-trade cap (its premium is above ~Rs.256). This is by design — don't override it lightly (see "Position sizing" below)
- > monthly loss limit hit (MTD Rs.-15200, limit Rs.15000) — the month's realized loss reached the cap; no new trades until next month

The alert still fires — you just don't get the button.

"★ HIGH CONVICTION" tag (V5 → now a gate for UP)

Alerts carry a ★ HIGH CONVICTION tag and a VA: POC.. / VAH.. / VAL.. line showing the day's value area. Clearing the value area used to be just a label; as of V7 it's a required filter — an UP breakout must close above VAH (and DOWN below VAL) to fire at all. So every UP alert you receive is high-conviction by construction (the ★ will be there). DOWN alerts now carry the opposite marker — (!) LOW CONVICTION — by construction: they pass the same shape of gate but the PE side is not the validated edge, so it's surfaced for observation, never as a real-money recommendation.

Important: the star does not mean "trade bigger" — size is always one lot. It's a quality marker, not a sizing signal.

"TRACKING ONLY" (weekly, early in the cycle)

Earlier in the monthly cycle (monthly more than 15 days out) you'll see alerts like:

```
Expiry: WEEKLY · monthly 22d out -- TRACKING ONLY (no order)
ITM-1 CE: NIFTY25JUN24650CE
LIMIT: Rs.138.00 | Qty: 65
[TRACKING ONLY -- weekly journaled as paper, not actionable]
```

There is no button and nothing to do. The bot records it as a *paper* trade so the weekly half's performance can be compared later. These never use real money — not even in LIVE mode. You act only on ACTIONABLE (monthly) alerts. This is normal for roughly the first two weeks after each monthly expiry.

Live fill messages (V5 — only in LIVE mode)

In LIVE mode the bot confirms your order actually filled before acting. You may see:

- > "BUY NOT FILLED" — price moved past your limit before the order filled, so no position was taken. Nothing to do; wait for the next signal. (This is a safety feature, not an error.)
- > "(!) PARTIAL FILL" — only part of the lot filled (rare). The message tells you how many; check Kite.
- > "SELL NOT FILLED ... retrying" — an exit didn't fill on the first try; the bot keeps trying. The broker's auto-square-off (~3:20 PM) is the final backstop.

"Market closed. ORB monitor stopping."

End of day at 3:30 PM. Shows total fires.

Your trading checklist (still applies)

The button tap places the entry. Stops, exits, sizing — still your job:

1. Define stop-loss before the position fills. Standard rule: stop on the option = 30–40% of premium paid. On the futures level = re-entry into ORB range (use a Kite GTT for this). 2. Position size sanity. 1 lot of Nifty at premium Rs.150 = Rs.9,750 capital deployed. Stop at 30% = Rs.2,925 risk. That should be $\leq 1\text{--}2\%$ of trading capital. If it's more, don't tap the button — the bot doesn't know your account size. 3. Set a target before the trade goes anywhere. Common: $1.5\times\text{--}2\times$ the risk. 4. Don't average down if it goes against you.

Daily shutdown (after market close)

The bot auto-sends Market closed. ORB monitor stopping. at 3:30 PM and the main loop exits. But the Python process may still be alive for a few seconds (the Telegram listener thread is a daemon and dies with the process — but only when the main thread exits cleanly).

Verify and clean up:

```
ssh ubuntu@<lightsail-ip>
ps aux | grep orb_monitor | grep -v grep
```

If you see a line, kill it:

```
pkill -f orb_monitor.py
```

Then verify it's gone:

```
ps aux | grep orb_monitor | grep -v grep
```

Empty output = clean.

Why this matters: if you start tomorrow's session without killing today's process, you'll have two bots running, doubled alerts, and possibly two BUY orders on the same button tap. Always kill before starting fresh.

Daily P&L review (V3)

Every paper / live trade is journaled to trades.db (a SQLite file in ~/nifty-orb-alerts/). One row per position with entry, exit, and P&L.

Quick review with report.py

```
cd ~/nifty-orb-alerts && source venv/bin/activate
python3 report.py # today
python3 report.py 2026-05-08 # specific date
python3 report.py 2026-05-08 --mode PAPER # filter by mode
```

Output is a per-trade table plus daily totals (wins / losses / win rate / net P&L).

Correcting a row (rare)

Back up first, then UPDATE. pnl is stored, not computed on read — always recompute it whenever you change entry_price or exit_price, or the row will be inconsistent:

```
cp trades.db trades.db.bak.$(date +%Y%m%d-%H%M)

sqlite3 trades.db "
BEGIN;
UPDATE trades
SET entry_price = <NEW_ENTRY>,
    pnl         = (exit_price - <NEW_ENTRY>) * qty
WHERE id = <ID>;
COMMIT;
"
```

Reconstructing a trade from orb.log

V3 writes structured, timestamped logs for every signal evaluation, fire, fill, position poll, and exit. After a trading day you can fully reconstruct any trade:

```
grep SIGNAL_FIRED orb.log # all signals that fired today
grep "EXIT reason" orb.log # all exits with realized P&L
grep position_tick orb.log # minute-by-minute monitoring while holding
grep "tick candle" orb.log # every 5-min signal evaluation (including skips)
```

Useful when a trade behaved unexpectedly — you'll see the exact ORB levels, volume ratio, VWAP, and signal decision for each candle.

Tuning the filters

All thresholds live in config.py on the Lightsail box. Edit, save, restart.

Setting	Default	What it does	When to change
VOLUME_MULTIPLIER	1.2	Breakout vol must be $\geq$ this $\times$ the 20-candle avg	Up to 1.5 for stricter; down to 1.0 if missing valid moves
LOOKBACK_CANDLES	20	Number of recent 5-min candles for the volume average	Rarely needs changing
MAX_FIRES_PER_DIRECTION	2	Max alerts per direction per day	1 for ultra-conservative; 3 if you want every clean re-break

BREAKOUT_BUFFER	10	Pts beyond ORB needed to count as a breakout	Higher = fewer fakeouts but later entries
REARM_BUFFER	15	Pts back inside ORB needed before re-arm	Higher = stricter re-entry filter
LOT_SIZE	65	Nifty lot size (NSE-defined)	Update if NSE changes the lot size
LIMIT_BUFFER	0.50	Rs. above best ask for the BUY limit	Higher = more fill certainty, slightly worse price
MAX_SPREAD_PCT	0.05	Skip auto-order if (ask-bid)/mid > this	Lower for stricter liquidity filter
MAX_PREMIUM	500	Skip auto-order if option premium > this	Catches expensive far-from-spot options
ORDER_PRODUCT	"MIS"	MIS = intraday auto-squareoff	Use "NRML" only if carrying overnight
DRY_RUN	True	Log instead of placing real orders	Flip to False after one clean DRY_RUN day
PAPER_TRADE	True	Simulate fills + auto-exits + journal to trades.db (no real orders)	Flip to False once 2 weeks of paper history look profitable

After editing config.py:

```

pkill -f orb_monitor.py
nohup python orb_monitor.py > orb.log 2>&1 &

```

Troubleshooting

"ORB monitor started" never arrives

```
tail -100 orb.log
```

Common errors:

Error contains	Fix
No valid access token for today	You forgot python auth.py. Run it.
TokenException	Token invalid/expired — usually because you logged in before the ~7:30 AM reset. Re-run auth.py *after ~7:45 AM* and restart.
connection, timeout, network	Lightsail momentarily lost network. Restart.
No active Nifty futures found	Holiday or expiry edge case. Check NSE calendar.

Telegram alerts not arriving

```
python -c "from telegram_alert import send_alert; send_alert('test')"
```

- > Nothing on phone → bot token wrong, or you blocked the bot, or chat ID wrong.
- > Get Telegram send failed: ... → the error tells you.

Tapped the button but no order/confirmation came back

1. Check orb.log — look for Callback handler error: or ORDER FAILED. 2. Most likely cause: the access token was invalidated — almost always because you logged in before the ~7:30 AM Kite reset (the token then dies at the reset). Re-run auth.py *after ~7:45 AM* and restart. The button you already tapped won't be retried — place that one manually.
3. Could also be that the bot process died. `ps aux | grep orb_monitor | grep -v grep` — if empty, restart.

"Auto-order skipped" on every alert

Spread or premium guards are too tight. Loosen MAX_SPREAD_PCT (e.g., to 0.07) or raise MAX_PREMIUM (e.g., to 800). Restart.

Process still running from yesterday

```
ps aux | grep orb_monitor | grep -v grep
pkill -f orb_monitor.py
```

Got two identical alerts within seconds

Two bots running. `pkill -f orb_monitor.py` (kills both), then start fresh.

Things you should NOT do

- > Don't flip DRY_RUN = False without one full day of dry-run validation. Real money, irreversible orders.
- > Don't run python auth.py after market open. The bot may already be exiting.
- > Don't push config.py to GitHub. It's in .gitignore. Secrets stay on Lightsail.
- > Don't paste your Kite API secret, access token, or TOTP into chats, screenshots, or anywhere outside config.py. If you do — rotate immediately at <https://developers.kite.trade/apps>.
- > Don't tap a button after the alert is more than ~30 seconds old — the underlying option price has moved and the LIMIT may now be too far below market to fill.
- > Don't let one losing day double your size the next day to "recover".

Risk discipline (read once a week)

- > Options decay daily. A 50% drop in the option premium is normal and frequent.
- > Win rate will not be 100%. Expect 40–55% winners. Profitability comes from average winner > average loser.
- > Daily loss cap: stop trading after 2 consecutive losses or 3% of capital, whichever comes first.
- > Weekly review: Sunday evening, review every alert vs. trades you took vs. P&L. Look for patterns (wide ORBs lose, tight ORBs win, etc.).

Optional: auto-restart with systemd

If you want the bot to survive a Lightsail VM reboot without you SSHing in to start it manually, create a systemd service. You still need to refresh the Kite token every morning — systemd just handles the process lifecycle.

One-time setup on the VM:

```

sudo tee /etc/systemd/system/orb-monitor.service > /dev/null <<'EOF'
[Unit]
Description=Nifty ORB Monitor
After=network-online.target

[Service]
Type=simple
User=ubuntu
WorkingDirectory=/home/ubuntu/nifty-orb-alerts
ExecStart=/home/ubuntu/nifty-orb-alerts/venv/bin/python orb_monitor.py
Restart=on-failure
RestartSec=30
StandardOutput=append:/home/ubuntu/nifty-orb-alerts/orb.log
StandardError=append:/home/ubuntu/nifty-orb-alerts/orb.log

[Install]
WantedBy=multi-user.target
EOF

sudo systemctl daemon-reload
sudo systemctl enable orb-monitor

```

Daily morning then becomes:

```

python auth.py                # refresh token (still manual)
sudo systemctl restart orb-monitor  # picks up the new token

```

Daily shutdown:

```
sudo systemctl stop orb-monitor
```

Status check:

```
sudo systemctl status orb-monitor
```

If you're not comfortable with systemd, stick with nohup — it works fine for daily manual operation.

Roadmap

Version	Status	What it adds
V1	Done	Telegram alerts only
V1.5	Done	Re-arm logic — up to 2 fires per direction
V2	Done	5-min tracking; ITM-1 strike; 1-tap LIMIT order with safety guards; DRY_RUN flag

V3	Live now	PAPER_TRADE mode; auto SL/target/timeout exits; trades.db journal at alert-time price; structured orb.log; report.py daily summary
V4	Done	Risk-based position sizing; monthly loss limit
V5	Done	Live fill-confirmation; VAH/VAL high-conviction tagging
V6	Done	Trailing-stop exit ("let winners run"); USE_TRAILING_STOP flag
V7	Live (paper)	DTE expiry routing (monthly ≤ 15 DTE actionable / weekly early-cycle paper-tracked) + VAH gate + put-side support-OI gate — the full validated strategy. DTE_MONTHLY_MAX, WEEKLY_TRACK_ONLY. Paper-validating before live flip
V8	Live (paper)	DOWN/PE mirror gate: VAL + resistance-call OI rising since 9:30. PE flagged LOW conviction (not a validated edge — noise filter + honest flag, paper only)
Later	Maybe	Multi-symbol (BankNifty, FinNifty); holiday calendar; web dashboard with live P&L

Reference

- > Repo: <https://github.com/anilkumarjonnalagadda/nifty-orb-alerts> (private)
- > Kite developer console: <https://developers.kite.trade/apps>
- > NSE holiday calendar: <https://www.nseindia.com/resources/exchange-communication-holidays>
- > Bot logs on Lightsail: `~/nifty-orb-alerts/orb.log`
- > Cron logs (token reminder): `~/nifty-orb-alerts/cron.log`